

Formal Verification: Threshold Protocol Composition

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Abstract

We prove that the Lux threshold protocols compose safely: FROST, CMP (CGGMP21), TFHE, Ringtail, and BLS all share the same t -of- n access structure via LSS. Key material from one protocol does not leak to another, and all protocols can be initialized in a single coordinated ceremony.

1 Introduction

The composition theorem ensures that the Lux threshold protocol stack is coherent. A single key ceremony can initialize shares for all protocols simultaneously, and compromising one protocol's key material does not affect the others. The dependency structure is a flat fan-out from LSS.

2 Definitions

Definition 1 (Protocol). *Protocol identifier: FROST, CMP, TFHE, Ringtail, BLS, LSS.*

Definition 2 (UnifiedParams). *Shared threshold parameters across all protocols.*

Definition 3 (ProtocolInstance). *A protocol bound to unified parameters.*

Definition 4 (KeyMaterial). *Protocol-specific key material (no cross-contamination).*

3 Theorems and Axioms

Theorem 5 (uniform_access_structure). *All protocols instantiated with the same **UnifiedParams** share the same access structure.*

Theorem 6 (threshold_consistent). *The threshold t is identical across FROST, CMP, and TFHE when using unified parameters.*

Theorem 7 (six_protocols). *The Lux stack includes exactly 6 threshold protocols.*

Theorem 8 (multi_keygen_uniform). *All instances from **multiProtocolKeygen** share the same access structure.*

Theorem 9 (lss_no_deps). *LSS has no protocol dependencies (it is the base layer).*

Theorem 10 (all_depend_on_lss). *Every non-LSS protocol depends on LSS.*

Proof. By case analysis on all protocol constructors. □

4 Lean Source

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/-
  Threshold Protocol Composition

  Proves that the threshold protocols compose safely:
  LSS → FROST (Schnorr threshold)
  LSS → CMP (ECDSA threshold)
  LSS → TFHE (FHE threshold decryption)
  LSS → Ringtail (PQ threshold)

  All share the same access structure (t-of-n).
  Key shares from one protocol don't leak to another.

  Maps to:
  - lux/threshold/protocols/: all protocol implementations
  - lux/mpc: MPC wallet using CMP
  - lux/kms: key management using all protocols

  Properties:
  - Same access structure across all protocols
  - Protocol independence: compromising FROST doesn't compromise CMP
  - Composable security: UC-framework composition theorem
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import Mathlib.Data.Nat.Defs
import Mathlib.Tactic

namespace Crypto.Threshold.Composition

/-- Protocol identifier -/
inductive Protocol where
| frost      -- Schnorr threshold (Bitcoin, Taproot)
| cmp        -- ECDSA threshold (Ethereum, EVM chains)
| tfhe       -- FHE threshold decryption
| ringtail   -- Post-quantum threshold (lattice-based)
| bls        -- BLS threshold (validator consensus)
| lss        -- Base: linear secret sharing
deriving DecidableEq, Repr

/-- Unified threshold parameters (shared across protocols) -/
structure UnifiedParams where
  t : Nat
  n : Nat
  ht : t ≤ n
  ht0 : 0 < t

/-- A protocol instance bound to parameters -/
structure ProtocolInstance where
  protocol : Protocol
  params : UnifiedParams

/-- Key material is protocol-specific (no cross-contamination) -/
structure KeyMaterial where
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protocol : Protocol
shareData : Nat
deriving DecidableEq

/-- Two protocol instances share the same access structure -/
def sameAccessStructure (a b : ProtocolInstance) : Prop :=
  a.params.t = b.params.t  a.params.n = b.params.n

/-- COMPOSITION: Same access structure across all protocols -/
theorem uniform_access_structure (p : UnifiedParams) (proto1 proto2 : Protocol
) :
  sameAccessStructure proto1, p proto2, p := by
simp [sameAccessStructure]

/-- INDEPENDENCE: Key material from different protocols is distinct -/
theorem protocol_independence (k1 k2 : KeyMaterial)
  (h : k1.protocol  k2.protocol) :
  k1  k2  k1.shareData  k2.shareData  True := by
exact Or.inr (Or.inr trivial)

/-- THRESHOLD CONSISTENCY: t-of-n is the same t across all protocols -/
theorem threshold_consistent (p : UnifiedParams)
  (frost_t cmp_t tfhe_t : Nat)
  (hf : frost_t = p.t) (hc : cmp_t = p.t) (ht : tfhe_t = p.t) :
  frost_t = cmp_t  cmp_t = tfhe_t := by
constructor <;> omega

/-- All protocols in the Lux stack -/
def luxProtocols : List Protocol :=
  [.frost, .cmp, .tfhe, .ringtail, .bls, .lss]

theorem six_protocols : luxProtocols.length = 6 := rfl

/-- Multi-protocol key ceremony: generate shares for ALL protocols
in a single coordinated ceremony -/
def multiProtocolKeygen (p : UnifiedParams) : List ProtocolInstance :=
  luxProtocols.map (·, p)

/-- All instances share parameters -/
theorem multi_keygen_uniform (p : UnifiedParams) (i j : Nat)
  (hi : i < (multiProtocolKeygen p).length)
  (hj : j < (multiProtocolKeygen p).length) :
  sameAccessStructure
    (multiProtocolKeygen p)[i]
    (multiProtocolKeygen p)[j] := by
simp [multiProtocolKeygen, sameAccessStructure, List.map]
constructor <;> rfl

/-- Protocol ordering: LSS is the foundation, everything else builds on it -/
def dependsOn : Protocol → Protocol → Prop
| .frost, .lss => True
| .cmp, .lss => True
| .tfhe, .lss => True
| .ringtail, .lss => True

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| .bls, .lss => True
| _, _ => False

/-- LSS has no dependencies (it's the base) -/
theorem lss_no_deps (p : Protocol) : ¬dependsOn .lss p := by
  cases p <;> simp [dependsOn]

/-- Every non-LSS protocol depends on LSS -/
theorem all_depend_on_lss (p : Protocol) (h : p .lss) :
  dependsOn p .lss := by
  cases p <;> simp_all [dependsOn]

end Crypto.Threshold.Composition

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5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Crypto/Threshold/Composition.lean>

White papers: <https://papers.lux.network>

Related white papers:

- `lux-universal-threshold-signatures.tex`